

# VEER ARORA

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## PROFILE

Backend and AI engineer building systems that report their own reliability: evaluation harnesses with held-out splits, policy gates on agent actions, and retrieval that cites its source. Validated an 8-model production MLOps platform at L&T Finance on GCP and Kubeflow. Recent work includes an LLM support agent whose every write passes a policy and escalation gate, and a RAG system where recall@5 went from 0.37 to 0.77 against a measured 0.85 ceiling — measured, not estimated. B.Tech Computer Science & Design, 2026.

## SKILLS

**AI & LLM** LLM tool-calling agents, RAG, LangGraph, evaluation harnesses (tool-selection, task-success, grounding, cost/latency), OpenAI / Anthropic / Google Gemini APIs  
**Backend & Web** FastAPI, PostgreSQL, row-level security, SQLAlchemy, Alembic, REST APIs, GitHub Actions CI, pytest, Docker, Git, React, TypeScript  
**Languages & Data** Python, SQL, Pandas, NumPy, data cleaning & validation, EDA  
**Cloud & MLOps** Google Cloud Platform, Kubeflow, pipeline testing, UAT  
**Analytics & ML** correlation analysis, descriptive statistics, outlier & anomaly detection, scikit-learn  
**Reporting & Tools** Power BI, Matplotlib, Seaborn, PDF report generation, Streamlit, SQLite, n8n, OpenCV

## EXPERIENCE

### L&T Finance — Data Analyst Intern

Jun 2025 – Aug 2025

- Validated Nostradamus, an 8-model MLOps platform on GCP and Kubeflow spanning EWS, Banking, Self-Cure and Collections workflows, ahead of production release.
- Caught an output-formatting defect in the EWS model that left its output column blank downstream — a failure that shows up as missing data rather than as an error — and drove root-cause analysis with the developers and MLOps engineers who owned the pipeline.
- Wrote SQL verification across pipeline stages: data-integrity and row-level checks that separated “the pipeline ran” from “the pipeline produced the right numbers”.
- Ran end-to-end pipeline validation and UAT in Python and SQL — test planning, defect tracking, deployment-readiness review.

### E-Cell, IIT Bombay — Campus Executive

Aug 2023 – Feb 2024

- Promoted entrepreneurship programs, workshops and competitions; coordinated outreach, event logistics and mentor coordination across the student community.

## PROJECTS

### AgentOps — AI Support Engineer

[github.com/veer0608/agentops](https://github.com/veer0608/agentops)

Python · FastAPI · LangGraph · SQLAlchemy · Alembic

- Built an AI support agent that takes real actions (create/update tickets, escalate) behind a policy gate that auto-executes, routes to a human review queue, or blocks each action on business rules and confidence thresholds; writes are idempotent and recorded in an append-only audit log.
- Made the model provider-agnostic — OpenAI, Anthropic and Google Gemini swappable via one config value — and implemented a second agent on LangGraph behind the same interface, using the eval harness to prove both implementations score identically.
- Wrote a scenario-based evaluation harness scoring six dimensions (tool-selection precision/recall/F1, task-success, escalation accuracy, citation-grounding, cost, latency), persisting every run for before/after tracking.
- FastAPI + SQLAlchemy + Alembic (SQLite dev / Postgres); 33 automated tests; read tools include RAG-backed doc search and GitHub-issue lookup.

### CiteRAG — Retrieval-Augmented QA with Golden-Set Eval

[github.com/veer0608/citerag](https://github.com/veer0608/citerag)

Python · FastAPI · SQLite-vec · SQLAlchemy

- Built a retrieval-augmented QA system over messy 10-K PDFs where every answer is grounded in cited source chunks with page numbers.
- Created a golden-set eval harness (recall@k, MRR) and lifted recall@5 from 0.37 to 0.77 against a measured ceiling of 0.85, by iterating on chunking strategy and adding a cross-encoder re-ranker — measured, not guessed.
- Local embeddings and reranker on SQLite-vec; the retrieval path and the entire eval harness run with no paid API.

### schemablind — Schema-Blind SQL Agent

[github.com/veer0608/schemablind](https://github.com/veer0608/schemablind)

Python · SQLite · BIRD benchmark · pytest

- Built a SQL agent given no schema: it explores a database it has never seen through four read-only verbs — list tables, describe, sample rows, run query — and composes its own answer, including learning from its own failed queries.
- Scored on BIRD execution accuracy: the agent's rows against the reference query's rows on the same database, with no judge model deciding whether SQL “looks right”.

- Held 40% of questions back as a test split assigned by hash of the question text, so a prompt tuned on failures cannot choose what it is marked on.
- Proved the harness before trusting it — an oracle solver must score 100% and a mute solver 0%, asserted in CI on every push. Sandboxed model-composed SQL behind four independent layers, with CI asserting the database is byte-identical after attempted writes. 119 tests.

### AgencyDesk — Multi-Tenant SaaS Platform

[github.com/veer0608/agencydesk](https://github.com/veer0608/agencydesk)

*React · FastAPI · PostgreSQL · Docker*

- Built a multi-tenant client/project management platform where one deployment serves many isolated agencies, each with its own clients, projects and client portal.
- Enforced tenant isolation and internal-vs-client visibility with PostgreSQL row-level security and composite foreign keys, so cross-tenant access is impossible even if application code is wrong; the API connects as an unprivileged database role.
- Wrote 70 automated tests proving cross-tenant isolation and role-based visibility; containerized with Docker Compose and added GitHub Actions CI that builds and tests on every push.

## EDUCATION

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### Madhav Institute of Technology & Science, Gwalior — B.Tech, Computer Science & Design

2022 – 2026

Shanti Niketan HS School, Vidisha — Class 12 (2021) · Trinity Convent Sr. Sec. School, Vidisha — Class 10 (2019)

## CERTIFICATIONS & ACTIVITIES

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Affective Computing — NPTEL (Jan–Apr 2024) · Digital Design with Verilog — NPTEL (Jan–Apr 2025) · Rational Thinking — Udemy  
R&D Coordinator, Startup Cell, MITS (Jan 2023 – Jun 2026) · Core Team Member, Digital Learning Group, MITS (Feb 2023 – Jul 2024) · State-level swimmer

Languages: English, Hindi